Industrial Internship Report on

" Prediction of Agriculture Crop Production in India"

Prepared by

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Executive Summary
This report provides details of my Industrial Internship conducted through upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). During the internship, I worked on a **Machine Learning project for Agriculture Crop Production Prediction**. The project focused on analyzing agricultural data and building a model to predict crop production using factors such as crop type, year, and area. This internship helped me gain practical experience in **Python, Machine Learning, data analysis, model evaluation, and Streamlit**. Overall, it was a valuable learning experience and helped me understand how a real-world ML project is developed. .

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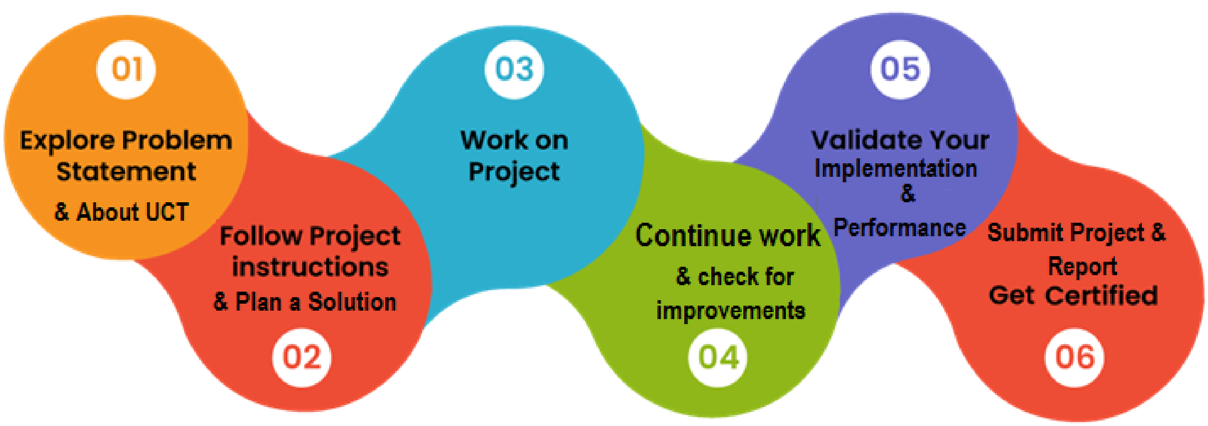
1 Preface

The six-week internship gave me an opportunity to work on a practical Machine Learning project. During the internship, I worked on **Agriculture Crop Production Prediction**, where I analyzed agricultural data and developed a model to predict crop production.

This internship was useful for my career development because it helped me apply the Machine Learning concepts I had learned in academics to a practical problem. I worked on different stages of the project, including data analysis, preprocessing, model training, evaluation, and prediction.

The project was provided as part of the internship opportunity through **upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT)**. The program was planned for six weeks, during which I worked on the project and prepared the required report and deliverables.



I sincerely thank my mentors, faculty members, and peers for their continuous support and guidance

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

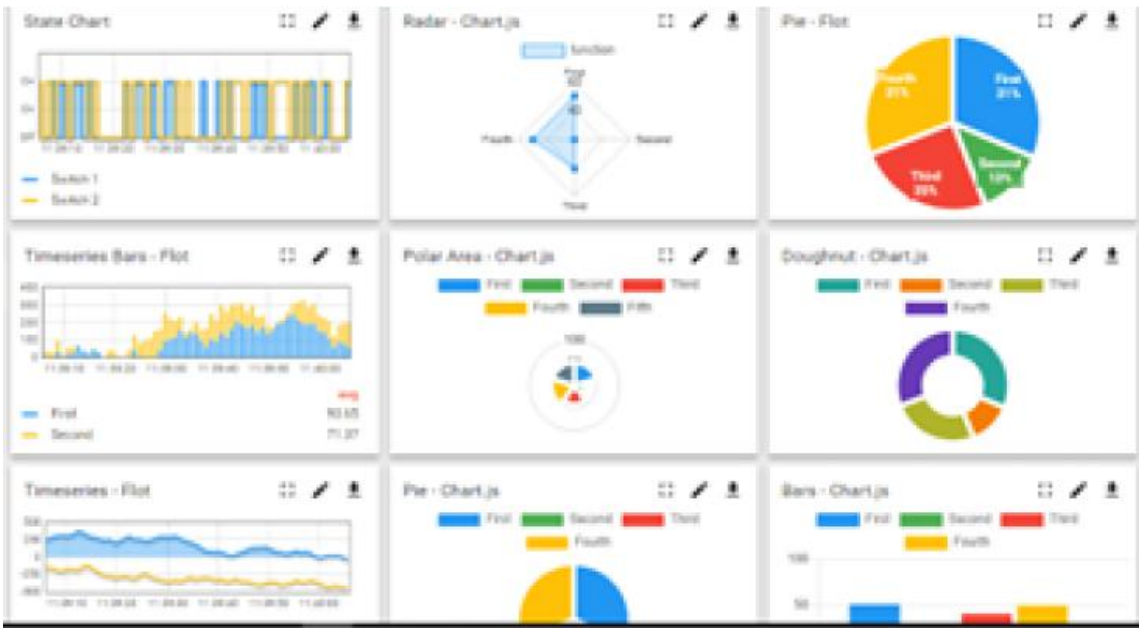
• Build Your own dashboard

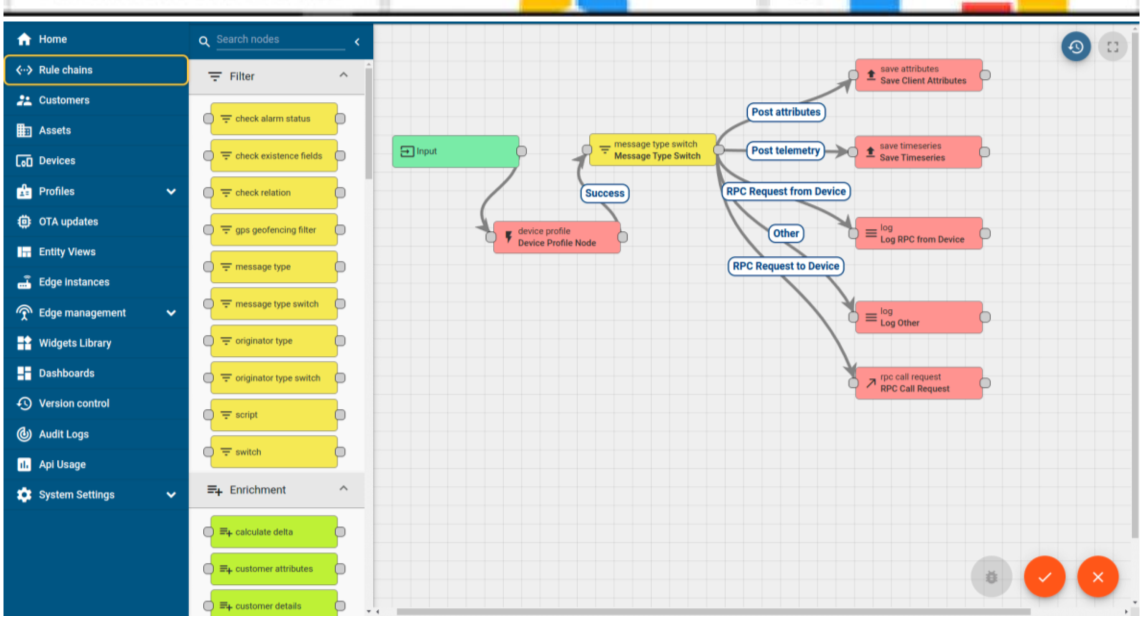
• Analytics and Reporting

• Alert and Notification

• Integration with third party application(Power BI, SAP, ERP)

• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

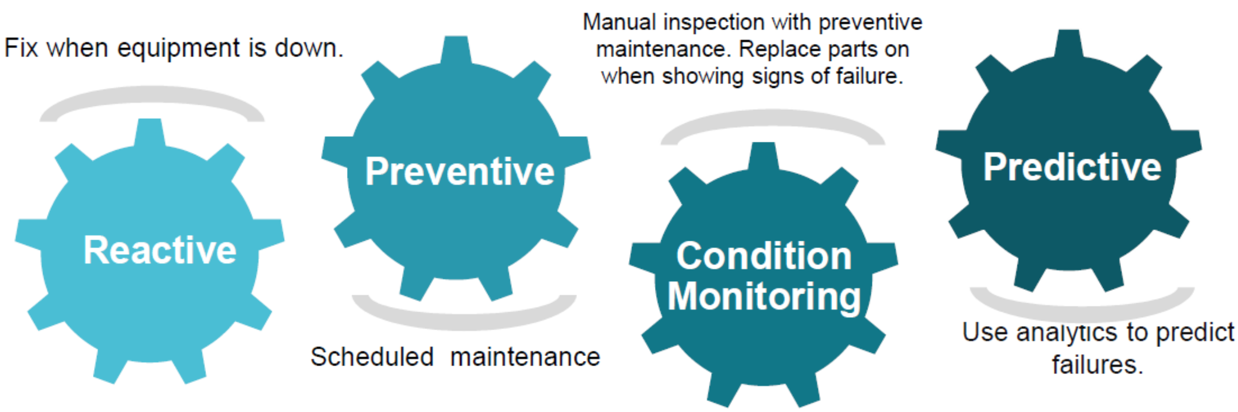
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

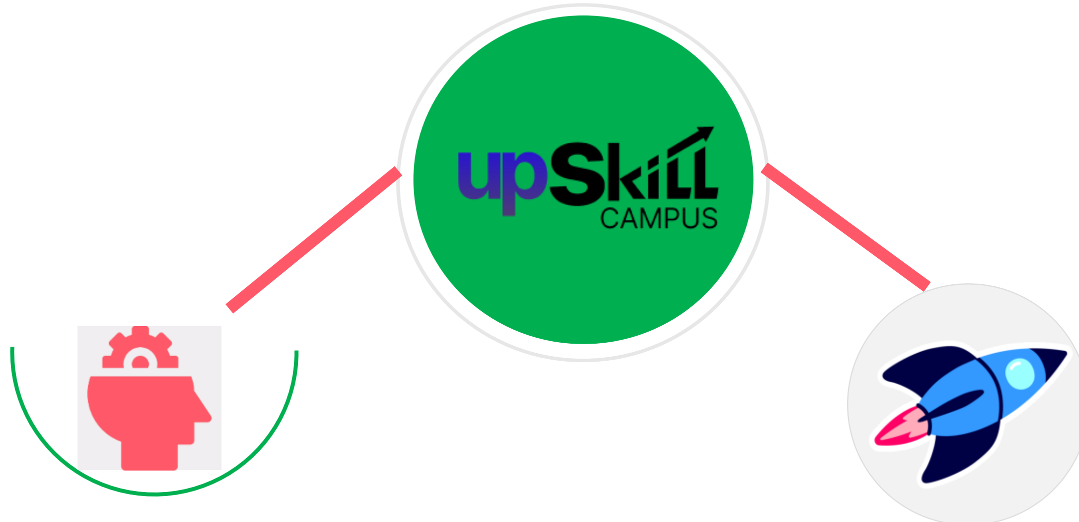
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

- **Career growth/upskilling**
  - Interview Preparation and skill building
  - upskilling Courses
  - Skill Assessment
  - Profile building
- **Professional networking**
  - Alumni Connections
  - Mentorship
  - Discussion/QA forum
- **Collaboration platform**
  - Project collaboration
  - Discussion forum
  - Tech updates
- **Job/internship platform**
  - Job portal
  - Internship portal
  - Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

3 Problem Statement

The assigned problem was to develop a Machine Learning based Agriculture Crop Production Prediction system. The main objective of the project was to use historical agricultural data to predict the production of different crops.

For this project, I worked with agricultural data containing information such as crop type, year, cultivated area, yield, and production. The data was first analyzed and prepared for Machine Learning. Different models such as Linear Regression, Decision Tree, and Random Forest were trained and compared based on their performance.

After evaluating the models, Linear Regression gave the best result for the selected features. The final model was then used to make crop production predictions based on the crop, year, and area provided by the user. A simple Streamlit web application was also developed to make the prediction system easier to use.

The project helped me understand how Machine Learning can be applied to an agricultural problem and how a model can be developed from raw data to a working application.

4 Existing and Proposed solution

4.1.1 Existing Solution

Agricultural production is generally estimated using historical data, statistical methods, and traditional analysis. These approaches can provide useful information, but they may require manual analysis and may not easily provide predictions for different crop and area combinations.

4.1.2 Proposed Solution

My proposed solution is a Machine Learning based Agriculture Crop Production Prediction **s**ystem. The system uses historical agricultural data containing information such as crop, year, and area to train a Machine Learning model. Different models were compared, and **Linear** Regression was selected based on its performance.

A Streamlit web application was also developed so that users can select a crop, enter the year and area, and get the predicted production.

4.1.3 Value Addition

The main value addition of this project is that it provides a simple and easy-to-use prediction **system** instead of requiring manual calculations. It also combines Machine Learning with a web interface, making the prediction process more accessible to users.

4.2 Code submission (Github link)

4.3 Report submission (Github link) :.

5 Proposed Design/ Model

The proposed system follows a simple Machine Learning workflow, starting from the agricultural dataset and ending with crop production prediction.

5.1.1 Design Flow

Agricultural Data → Data Analysis → Data Preprocessing → Feature Selection → Model Training → Model Comparison → Model Evaluation → Final Model → Prediction

5.1.2 1. Data Collection

The available agricultural datasets were collected and examined to understand the available crop, year, area, yield and production information.

5.1.3 2. Data Analysis and Pre-processing

The data was analyzed for its structure, missing values and required columns. The data was cleaned and converted into a suitable format for Machine Learning.

5.1.4 3. Feature Selection

For the final model, **Crop, Year and Area** were selected as input features, while **Production** was used as the target variable.

5.1.5 4. Model Training

Different Machine Learning algorithms were trained using the prepared dataset:

- Linear Regression
- Decision Tree
- Random Forest

5.1.6 5. Model Comparison and Evaluation

The models were compared using **MAE, RMSE and R^2 Score**. Linear Regression performed better than the other tested models.

5.1.7 6. Final Model

A Linear Regression model was selected as the final model and trained using a time-based approach. The final model achieved an **R^2 score of 0.9575** on the test data.

5.1.8 7. Prediction Application

The trained model and encoder were saved and connected to a **Streamlit web application**. The user can select a crop and enter the year and area to get the predicted crop production.

Agricultural Dataset

↓

Data Analysis

↓

Data Preprocessing

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Feature Selection

↓

Train ML Models

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Model Comparison

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Model Evaluation

↓

Linear Regression

↓

Save Final Model

↓

Streamlit Application

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Crop Production Prediction

6 Performance Test

The performance of the project was mainly evaluated based on the **prediction accuracy of the Machine Learning model**. Since this was a Machine Learning project, the main constraints considered were prediction error, model performance, and the ability of the model to work on unseen data.

The model was evaluated using **MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score**. To make the testing more realistic, a time-based test was also performed by using earlier years for training and a later year for testing.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Result
TC01	Valid crop, year and area	Production should be predicted
TC02	Different crop and area values	Model should provide a prediction
TC03	Unseen test data	Model should give reasonable results
TC04	Model performance evaluation	MAE, RMSE and R^2 should be calculated
TC05	Streamlit application	User should be able to enter details and get prediction

6.2 Test Procedure

First, the dataset was divided according to the year. The data from earlier years was used for training and the later year was used for testing. The Linear Regression model was trained using **Crop, Year and Area** as input features.

After training, the model predicted production values for the test data. These predicted values were compared with the actual production values using MAE, RMSE and R^2 Score.

The trained model was then connected to the Streamlit application and tested with different crop, year and area values.

6.3 Performance Outcome

The final Linear Regression model achieved the following results on the time-based test data:

Metric	Result
MAE	24.27
RMSE	47.51
R^2 Score	**0.9575**

The R^2 score of **0.9575** indicates that the model was able to explain a large portion of the variation in crop production in the test data.

The model was also successfully integrated with the Streamlit application. For example, when **Rice, 2010 and Area = 100** were provided as inputs, the application produced a predicted production value of **100.96**.

One limitation is that the model is based on the available historical dataset and only uses crop, year and area as final input features. Factors such as rainfall, temperature, soil quality, irrigation and weather conditions were not included. These factors can affect real-world crop production, so adding them in future versions could improve the model further.

7 My learnings

During this internship, I learned how to apply Machine Learning concepts to a practical problem. I improved my understanding of **Python, data analysis, data preprocessing, model training, model comparison, and model evaluation**.

I also learned how to compare different Machine Learning models and select a suitable model based on its performance. Developing the Streamlit application helped me understand how a trained model can be connected to a simple user interface.

This internship also improved my problem-solving and project development skills. These learnings will help me in my future career in **Machine Learning and AI** and will give me a better understanding of how to work on real-world projects.

8 Future work scope

- **More agricultural factors:** Rainfall, temperature, humidity, soil quality, irrigation and weather conditions can be added as input features.
- **Larger dataset:** More recent data from different regions can be included to improve the reliability of the model.
- **Advanced Machine Learning models:** Algorithms such as Gradient Boosting, XGBoost and other ensemble techniques can be tested and compared.
- **Hyperparameter tuning:** Different model parameters can be optimized to improve prediction performance.
- **Better data preprocessing:** More detailed handling of missing values, outliers and unusual data points can be implemented.
- **Regional prediction:** The system can be extended to predict production for specific states, districts or regions.
- **Data visualization:** Charts can be added to show crop production trends, area changes and yearly comparisons.
- **Online deployment:** The Streamlit application can be deployed online so that users can access it without running the project locally.
- **Regular data updates:** A system can be developed to automatically update the dataset with new agricultural information.
- **Improved user interface:** The application can be made more interactive and user-friendly with dashboards and detailed prediction reports.
- **Prediction for multiple crops:** The system can be extended to provide predictions for multiple crops at the same time.
- **Future forecasting:** The project can be extended to forecast crop production for future years using historical trends and additional environmental data.